

Transfer Learning for Olfactory Object Detection

Mathias Zinnen¹, Prathmesh Madhu¹, Peter Bell², Andreas Maier¹ and
Vincent Christlein¹

¹Pattern Recognition Lab, Friedrich-Alexander-Universität Erlangen-Nürnberg
(FAU), 91058 Erlangen, Germany

²Germanistik und Kunstwissenschaften (Fb09), Philipps-Universität Marburg,
35032 Marburg, Germany

1 Introduction

Smells are an important, yet overlooked part of cultural heritage [1]. The Odeuropa project¹ analyzes large amounts of visual and textual corpora to investigate the cultural dimensions of smell in 16th – 20th century Europe. The study of pictorial representations bears a specific challenge: the substrate of smell is usually invisible [10].

Object detection is a well-researched computer vision technique, and so we start with the recognition of objects, which may then serve as a basis for the indirect recognition of more complex, and possibly more meaningful, smell references such as gestures, spaces, or iconographic allusions [16].

However, the detection of olfactory objects in historical artworks is a challenging task. The visual representation of objects differs significantly between artworks and photographs [5]. Since state-of-the-art object detection algorithms are trained and evaluated on large-scale photographic datasets such as ImageNet [13], MS COCO [7], or OpenImages [6], their performance drops significantly when applied to artistic data. This *domain gap* between standard object detection datasets and artistic imagery can be mitigated by training directly on artworks, either by using existing datasets or by creating an annotated dataset for the target domain. Another challenge is the mismatch between object categories present in modern datasets and historical olfactory objects, caused by historical diachrony on the one hand [9], and the particularity of some smell-relevant objects on the other [16], [2].

2 Methodology

To overcome the domain gap and category mismatch between our application and the existing datasets, we apply transfer learning – a training strategy

¹www.odeuropa.eu



3. *Quamvis floriferus sit gratus naribus hortus,
Sepe tamen dulci fel sub odore latet*

Figure 1: *Smell*. The Five Senses. 1558 – 1617. Jan Pietersz Saenredam. National Gallery of Art. Public Domain

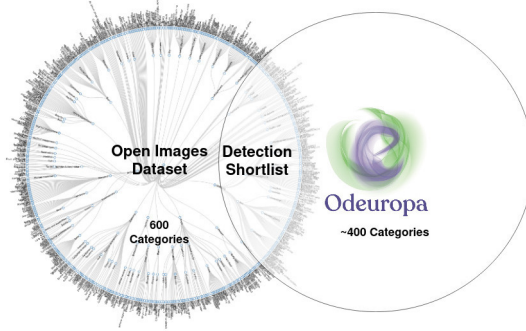


Figure 2: Category overlap between Odeuropa & OpenImages categories

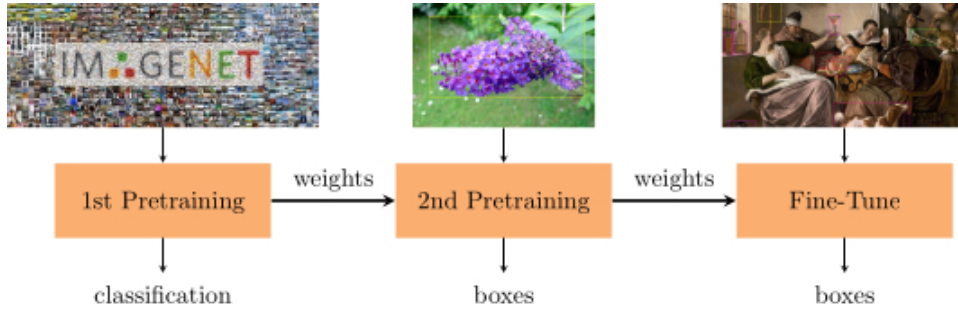


Figure 3: Transfer learning training strategy illustration. We start with a backbone pre-trained on ImageNet for classification, use this model to train an object detection system using different datasets. Finally, the object detection model is fine-tuned on the olfactory artworks dataset.

where machine learning algorithms are pre-trained in one domain and then fine-tuned in another, greatly decreasing the amount of required training data in the target domain ([11],[15],[8]).

We are continuously collecting and annotating artworks with possible olfactory relevance from multiple museum collections. Based on these, we created a dataset of olfactory artworks containing 16 728 annotations on 2 229 artworks. From this full set of annotations, we created a test set of 3 416 annotations on 473 artworks, while the remaining data was used for training.

A common transfer learning procedure is to use detection backbones that have been pre-trained on ImageNet and fine-tune them for object detection [15]. We expand this strategy by an additional pre-training step, where we train an ImageNet pre-trained object detection network [12] using different datasets. Finally, we fine-tune the resulting model using our olfactory artworks dataset (fig. 3).

For pre-training, we use three different datasets, deviating to varying amounts from our olfactory artworks dataset in terms of categories and style

Table 1: An overview of domain & category similarity of the experiment datasets to our olfactory artworks

Dataset	domain similarity	category similarity	# categories
OpenImages	low	complete match	29
IconArt	high	medium	10
PeopleArt	medium	low	1

Table 2: Evaluation of object detection performance. The best performing model pre-trained with OI achieves an improvement of 6.5% pascal VOC mAP, and 3.4% COCO mAP over the baseline method without intermediate training. We report the evaluation for each pre-training dataset, averaged over five models, fine-tuned for 50 epochs on our olfactory artworks datasets. Best evaluation results are highlighted in bold. The merge of two datasets D_1 and D_2 is written as $D_1 \cup D_2$.

Pretraining Dataset	Pascal mAP(%)	COCO mAP(%)
None (Baseline)	16.8(± 1.3)	8.4(± 0.4)
OI	23.3(± 0.5)	11.8(± 0.4)
IA	22.6(± 1.2)	10.9(± 0.9)
PA	21.9(± 0.4)	10.5(± 0.2)
IA $\cup$ OI	21.8(± 0.1)	10.5(± 0.3)
IA $\cup$ PA	22.0(± 0.8)	10.6(± 0.3)
PA $\cup$ OI	22.6(± 0.3)	10.8(± 0.2)
OI $\cup$ IA $\cup$ PA	21.8(± 0.4)	10.5(± 0.2)

(table 1): a) Same Categories, Different Styles - A subset of OpenImages (OI) containing only odor objects results in a complete category match (fig. 2); however, since OpenImages contains only photographs, there is a considerable style difference. b) Different Categories, Same Styles - We apply two object detection datasets from the art domain, which are more similar in terms of style but contain different object categories, namely IconArt (IA) [4] and PeopleArt (PA) [14].

3 Results

To ensure a fair comparison between the different pre-training datasets, we reduce each of the datasets to the same size, train three models, and select the best according to a fixed validation set for each dataset. Additionally, we merge all three datasets, i. e., combining OI, IA, and PA, using the union over their respective classes. The resulting models are then fine-tuned on the

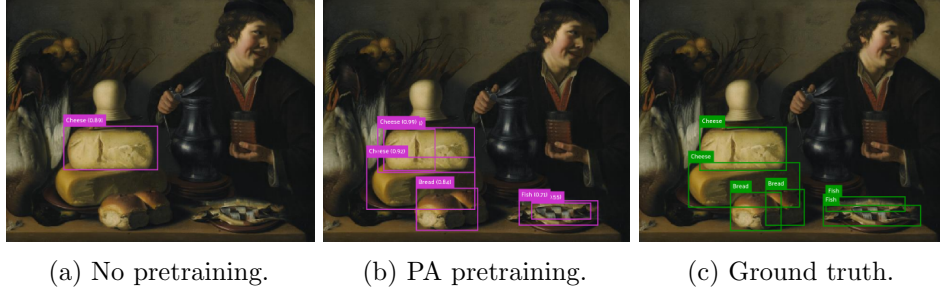


Figure 4: Exemplary object predictions for a detection model without intermediate training (a), with PeopleArt pretraining (b), and ground truth bounding boxes (c). Painting: *Boy holding a pewter tankard, by a still life of a duck, cheeses, bread and a herring*. 1625 – 1674. Gerard van Honthorst. RKD Digital Collection (<https://rkd.nl/explore/images/287165>). Public Domain.

training set of the olfactory artworks dataset and evaluated on a separate test set. To mitigate random variations that can occur during the training process, we train five separate models for each experimental setting and report their average. Evaluation results are reported in pascal VOC (mAP 50 [3]) and COCO mAP (mAP 50:95:5 [7]), the two standard metrics to evaluate object detection models. We conduct two separate sets of experiments: In the first, we fine-tune the whole network, including the backbone, to assess the detection performance under realistic conditions (table 2). We observe a performance increase for all used pre-training datasets, with an increase of 6.5%/3.4% boost in mAP 50 and COCO mAP, respectively, for the best performing pre-training scheme, which was achieved using the OI dataset. The exemplary object predictions in fig. 4 show that adding an additional pre-training stage can increase the number of recognized objects.

In a second set of experiments, we train only the detection head while the backbone remains frozen, to compare the quality of the intermediate representations that have been learned using the different pre-training schemes (table 3). While all pre-training schemes increase the performance, the relative increase for the OI dataset is remarkably higher. This suggests that the style similarity between the IA and PA datasets and our target dataset is less important than we expected. We can not yet conclude whether the superior performance of the OI dataset is due to the similarity in target categories. It could also be caused by other properties of the dataset. Further ablations, e.g., varying the set of OI categories are needed to more precisely assess the impact of category similarity on the detection performance, which we plan to conduct in a follow-up study. Interestingly, the performance of the merged datasets increases even in cases where OI is not part of the dataset merge. Given that we did not apply a sophisticated merging strategy,

Pretraining Dataset	mAP(%) 50	mAP(%) 50:95
None (Baseline)	11.7(± 0.2)	5.5(± 0.1)
OI	19.4(± 0.3)	9.5(± 0.1)
IA	13.8(± 0.4)	6.4(± 0.2)
PA	13.5(± 0.2)	6.7(± 0.1)
IA $\cup$ OI	16.0(± 0.3)	7.4(± 0.2)
IA $\cup$ PA	14.6(± 1.0)	6.7(± 0.5)
PA $\cup$ OI	15.8(± 0.7)	7.3(± 0.4)
OI $\cup$ IA $\cup$ PA	16.4(± 0.6)	7.6(± 0.2)

Table 3: Evaluation of object detection performance for fine-tuning of the detection heads only. All pre-training schemes increase the detection performance, while pre-training with OI leads to the best results with an increase of 7.7% mAP 50 or 4% COCO mAP. For every pre-training dataset, we report the evaluation averaged over five models, fine-tuned for 50 epochs on our olfactory artworks datasets each. Best evaluation results are marked in bold. The merge of two datasets D_1 and D_2 is written as $D_1 \cup D_2$.

the performance increase for training with merged datasets is encouraging. Developing strategies to improve the consistency of the merged dataset, e. g., weak labeling of categories not present in the respective merge partners, represents another promising line of future research.

We conclude that including an additional stage of object-detection pre-training can lead to a considerable increase in detection performance. While our experiments suggest that style similarities between pre-training and target dataset are less important than matching categories, further experiments are needed to verify this hypothesis.

4 Acknowledgements

We gratefully acknowledge the support of NVIDIA Corporation with the donation of the two Quadro RTX 8000 used for this research. The paper has received funding by Odeuropa EU H2020 project under grant agreement No. 101004469.

References

- [1] Cecilia Bembibre and Matija Strlič. Smell of heritage: a framework for the identification, analysis and archival of historic odours. *Heritage Science*, 5(1):1–11, 2017.
- [2] Sofia Collette Ehrich, Caro Verbeek, Mathias Zinnen, Lizzie Marx, Cecilia Bembibre, and Inger Leemans. Nose first: Towards an olfactory gaze for digital art history. *First International Workshop on Multisensory Data and Knowledge*, 2021. Online accessed, December 09, 2021.
- [3] Mark Everingham, Luc Van Gool, Christopher KI Williams, John Winn, and Andrew Zisserman. The pascal visual object classes (voc) challenge. *International journal of computer vision*, 88(2):303–338, 2010.
- [4] Nicolas Gonthier, Yann Gousseau, Said Ladjal, and Olivier Bonfait. Weakly supervised object detection in artworks. In *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, pages 0–0, 2018.
- [5] Peter Hall, Hongping Cai, Qi Wu, and Tadeo Corradi. Cross-depiction problem: Recognition and synthesis of photographs and artwork. *Computational Visual Media*, 1(2):91–103, 2015.
- [6] Alina Kuznetsova, Hassan Rom, Neil Alldrin, Jasper Uijlings, Ivan Krasin, Jordi Pont-Tuset, Shahab Kamali, Stefan Popov, Matteo Mallocci, Alexander Kolesnikov, et al. The open images dataset v4. *International Journal of Computer Vision*, 128(7):1956–1981, 2020.
- [7] Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *European conference on computer vision*, pages 740–755. Springer, 2014.
- [8] Prathmesh Madhu, Angel Villar-Corrales, Ronak Kosti, Torsten Bend-schus, Corinna Reinhardt, Peter Bell, Andreas Maier, and Vincent Christlein. Enhancing human pose estimation in ancient vase paintings via perceptually-grounded style transfer learning. *arXiv preprint arXiv:2012.05616*, 2020.
- [9] Maria-Cristina Marinescu, Artem Reshetnikov, and Joaquim Moré López. Improving object detection in paintings based on time contexts. In *2020 International Conference on Data Mining Workshops (ICDMW)*, pages 926–932. IEEE, 2020.
- [10] Lizzie Marx. Perfume and books of secret. *Exhibition Catalogue Mauritshuis, The Hague*, 2021.

- [11] Sinno Jialin Pan and Qiang Yang. A survey on transfer learning. *IEEE Transactions on knowledge and data engineering*, 22(10):1345–1359, 2009.
- [12] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in neural information processing systems*, 201, 2015.
- [13] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115(3):211–252, 2015.
- [14] Nicholas Westlake, Hongping Cai, and Peter Hall. Detecting people in artwork with cnns. In *European Conference on Computer Vision*, pages 825–841. Springer, 2016.
- [15] Fuzhen Zhuang, Zhiyuan Qi, Keyu Duan, Dongbo Xi, Yongchun Zhu, Hengshu Zhu, Hui Xiong, and Qing He. A comprehensive survey on transfer learning. *Proceedings of the IEEE*, 109(1):43–76, 2020.
- [16] Mathias Zinnen. How to see smells: Extracting olfactory references from artworks. In *Companion Proceedings of the Web Conference 2021*, pages 725–726, 2021.